Entrepreneurial Finance and Venture Capital

In class Test 1

March 9th, 2018

Name

_____

Number

WRITE YOUR NAME AND NUMBER IN <u>ALL PAGES</u>

This test consists of 4 questions.

You have <u>75 minutes</u>.

This is a closed book test. Calculators are allowed.

Good luck!

Group 1: Multiple choice (21 points)

(no explanation needed)

A. Which of the following actions would increase a firm's asset intensity? (7 points)

i. Paying suppliers within 30 days instead of 60
ii. Renting computing capacity from Amazon instead of buying servers
iii. Eliminating a free trial month for new subscribers
iv. None of the above

B. Which of the following statements are true? (7 points)

i. If asset intensity is *negative*, then a fast-growing, *profitable* firm may have *negative* free cash flow
ii. If asset intensity is *negative*, then a fast-growing, *unprofitable* firm may have *positive* cash-flow
iii. None of the above

C. Which of the following factors make staging investments in entrepreneurial finance valuable? (7 points)

i. Staging minimizes funding risk due to changes in the funding environment
ii. Staging maximizes the option value of abandoning unsuccessful projects
iii. None of the above

ANSWER:

A_____

B_____

C_____

Group 2 (7 points)

Suppose a firm is doubling its sales every year and that profitability and asset intensity are constant. If profitability is 5% and the firm is generating free cash flow of 0, what is its asset intensity?

Group 3 (2 points)

A new venture is seeking 7 million euros from a VC. The VC projects the company will be sold for 150 million euros in 4 years and is targeting an IRR of 40%. The company will not require additional funding. There are 2 million shares outstanding before the deal.

A. What equity stake will the VC demand? (6 points)

B. What are the pre and post-money valuation if the deal is done on these terms? (6 points)

C. How many shares does the VC get? What is the implied share price? (7 points)

Assume now that there will be one more funding round where a second VC takes a stake of 15%.

D. If the first VC correctly anticipates the need for the second round:

i. What equity stake will the first VC now demand? (6 points)

ii. What will be the post-money valuation of the first round? (6 points)

iii. What will be the share price of the first round? (6 points)

E. If the first VC does *not* anticipate the need for a second round, what IRR will he achieve if the company successfully exits for the projected 150 million? (7 points)

Group 4 (30 points)

You are planning to start a new venture to develop an innovative sensor for self-driving cars. To develop this technology you must invest 80 million euros, and there is only a 10% chance that you will succeed. This risk is entirely diversifiable, and the risk free rate is 0%. Developing the technology will take 5 years, and if you are successful it will be worth 750 million euros. If you fail it will be worth nothing.

A. What is the NPV of this project? (7 points)

B. Suppose that you can split the 100 million investment into two stages. You can first run an experiment that costs 10 million. With 50% probability, the outcome of the experiment is positive, and if you invest the remaining 70 million there is a 20% probability that you will develop the technology successfully. If the outcome of the experiment is negative then the technology will fail for sure. What is the value of the option to abandon the project after running the experiment? (7 points)

C. What if the experiment only has a 20% probability of having a positive outcome but the probability of successfully developing the technology if the outcome is positive is 50%? Is the option more or less valuable? You do not need to report the value, just explain your answer. (7 points)

D. A VC is considering investing the 10 million to fund the experiment. If they evaluate the investment using the VC method (that is, by discounting the success scenario), what discount rate should they use in order to achieve an *expected* return equal to the risk free rate of 0%? (7 points)